The Copy-Forward Pathway: A Fourth Axis of the Adaptive Window in Viability-Constrained Memory Systems

VCSM Theory Series — Paper 12

Abstract

Papers 10–11 identified two boundaries governing the adaptive window of the Viability-Constrained Coupled Map Lattice (VCML): a lower bound ν_{cryst} set by field decay, and an upper bound $\nu_{\max}$ set by consolidation access. Here we isolate a third structural maintenance pathway — copy-forward inheritance — which operates independently of both consolidation and decay. At birth, each new site inherits a blend of local noise and its neighbor's `fieldM`: $h_{\text{new}} = (1-\beta_s)\,\varepsilon + \beta_s F_j$, with β_s (`SEED_BETA`) controlling inheritance strength. We show that (i) ablating copy-forward ($\beta_s = 0$) has minimal impact at low ν where consolidation dominates, but (ii) strong copy-forward ($\beta_s = 0.50$) shifts ν^* by 4× into the mid-turnover regime. A product-collapse test ($\nu \times \beta_s = \text{const}$) fails: iso-product pairs vary in sg4 by up to 24×, because field quality F_{eq} itself degrades with ν. We conclude that the adaptive regime is governed by at least four independent parameters $\{\nu, \nu_{\text{cryst}}, \nu_{\max}^{\text{consol}}, \beta_s\}$ forming a viability volume in control space, not a single scalar law. This result motivates the 4D framing developed in Paper 13.

1 Background and Motivation

The VCSM learning rule (Papers 0–6) maintains spatial structure in a lattice of cells with mandatory turnover. Papers 8–11 progressively isolated the scalar parameters governing the *adaptive window* $[\nu_{\text{cryst}}, \nu_{\max}]$ of turnover rates ν at which spatial differentiation (sg4) peaks:

$$\nu_{\text{cryst}} = \frac{|\ln(\text{FIELD_DECAY})|}{\ln 2} \qquad \text{(lower bound: memory crystallizes below this)} \tag{1}$$

$$\nu_{\max}^{\text{consol}} \approx P_{\text{calm}} \cdot \alpha_F \cdot \text{FIELD_DECAY}^{1/\nu} \qquad \text{(upper bound: consolidation cannot keep up above this)} \tag{2}$$

However, three anomalies persisted across Papers 10–11: structure maintained at (i) WR = 7.2 where $P_{\text{calm}} \approx 0.001$, (ii) SS = 40 where $\nu_{\max}^{\text{consol}} < \nu_{\text{cryst}}$, and (iii) $\alpha_F = 0.04$ where the window was predicted to vanish. All three share a common candidate: the *copy-forward pathway*, in which cell births propagate existing field structure without requiring calm consolidation.

This paper directly tests the copy-forward pathway by sweeping `SEED_BETA` (β_s) from 0 (full ablation) to 0.50 (strong inheritance), and by testing whether the product $\nu \times \beta_s$ collapses the data as a simple rate model would predict.

2 Methods

2.1 Experiment 1: SEED_BETA Sweep

We hold all parameters at reference values (WR= 4.8, WD= 15, SS= 10, FIELD_DECAY= 0.9997, $\alpha_F = 0.16$, $\kappa = 0.020$) and vary $\beta_s \in \{0.00, 0.05, 0.10, 0.25, 0.50\}$ across the full 12-point ν grid (5 seeds per condition, 300 total runs).

At birth, the dead site's hidden state is reinitialized as:

$$h_{\text{new}} = (1 - \beta_s)\,\mathcal{N}(0, 0.1) + \beta_s\, F_j$$

where j is a random live neighbor. When $\beta_s = 0$, births inject pure noise; copy-forward is fully ablated.

2.2 Experiment 2: Product Collapse Test

We run a 4×4 grid: $\nu \in \{0.001, 0.002, 0.005, 0.010\}$ by $\beta_s \in \{0.05, 0.10, 0.25, 0.50\}$ (5 seeds, 80 total runs). If copy-forward rate $\Gamma_{\text{cf}} = \nu \times \beta_s \times F_{\text{eq}}$ with F_{eq} independent of ν, iso-product pairs $(\nu_1, \beta_{s,1})$, $(\nu_2, \beta_{s,2})$ with $\nu_1\beta_{s,1} = \nu_2\beta_{s,2}$ should yield equal sg4.

3 Results

3.1 Experiment 1: Copy-Forward Has Regime-Dependent Impact

Table 1: SEED_BETA sweep results. Columns show sg4 at each ν value. $\nu_{\text{cryst}} = 4.3 \times 10^{-4}$ fixed throughout.

β_s	ν^*	sg4*	sg4 profile at selected ν					
			10^{-4}	5×10^{-4}	10^{-3}	2×10^{-3}	5×10^{-3}	10^{-2}
0.00	0.0005	394	210	394	311	215	113	60
0.05	0.0005	402	212	402	324	233	121	64
0.10	0.0005	408	215	408	337	254	130	67
0.25	0.0005	419	223	419	376	328	161	80
0.50	0.0020	500	246	407	443	500	218	107

Three key observations from Table 1:

(1) Ablation at low ν is weak. At $\nu = 5 \times 10^{-4}$, sg4 changes from 394 ($\beta_s = 0$) to 419 ($\beta_s = 0.25$), a 6% increase. The consolidation pathway alone ($\beta_s = 0$) is nearly sufficient at low turnover. This confirms that copy-forward is not the primary mechanism in the frozen regime: long-lived sites consolidate via the standard VCSM pathway.

(2) Copy-forward dominates at moderate ν. At $\nu = 0.002$, sg4 rises from 215 ($\beta_s = 0$) to 500 ($\beta_s = 0.50$), a 133% increase. At this turnover rate, consolidation is limited by calm-streak timing ($P_{\text{calm}} \approx 0.015$), but copy-forward births propagate field structure without requiring calm periods.

(3) Strong copy-forward shifts ν^*. At $\beta_s = 0.50$, the optimal turnover rate shifts from $\nu^* = 5 \times 10^{-4}$ to $\nu^* = 0.002$, a 4× displacement. The copy-forward pathway effectively raises $\nu_{\max}$ by extending structural maintenance into the high-turnover regime.

Table 2: sg4 heatmap from the $\nu \times \beta_s$ grid. Products $\nu\beta_s$ are shown in parentheses. Iso-product pairs at $\nu\beta_s = 5 \times 10^{-4}$ (diagonal entries) vary from 19 to 443 — a 24× range.

$\nu \backslash \beta_s$	0.05	0.10	0.25	0.50
0.001	324 (5×10^{-5})	337 (10^{-4})	376 (2.5×10^{-4})	443 ($\mathbf{5\times10^{-4}}$)
0.002	233 (10^{-4})	254 (2×10^{-4})	328 ($\mathbf{5\times10^{-4}}$)	500 (10^{-3})
0.005	64 (2.5×10^{-4})	67 ($\mathbf{5\times10^{-4}}$)	80 (1.25×10^{-3})	107 (2.5×10^{-3})
0.010	19 ($\mathbf{5\times10^{-4}}$)	20 (10^{-3})	24 (2.5×10^{-3})	39 (5×10^{-3})

3.2 Experiment 2: Product Hypothesis Fails; F_quality is ν-Dependent

Table 2 shows the iso-product failure. The four **bold** entries all share $\nu\beta_s = 5 \times 10^{-4}$ but produce sg4 values of 443, 328, 67, and 19. The product hypothesis requires these to be equal; instead they span a 24× range.

The explanation is immediate from the field-norm proxy $\bar{F}$ (mean $\|F_i\|$ at end of run):

Table 3: Mean field norm $\bar{F}$ (proxy for F_{quality}) across the $\nu \times \beta_s$ grid. $\bar{F}$ decays faster than $1/\nu$ as ν increases, invalidating the constant-F_{quality} assumption.

$\nu \backslash \beta_s$	0.05	0.10	0.25	0.50
0.001	502	523	596	753
0.002	318	335	400	542
0.005	89	92	103	128
0.010	29	30	34	46

From $\nu = 0.001$ to $\nu = 0.010$ (10×), $\bar{F}$ at $\beta_s = 0.25$ drops from 596 to 34 (17.5×). Field quality therefore degrades *faster* than $1/\nu$. The copy-forward rate:

$$\Gamma_{\text{cf}} = \nu \times \beta_s \times F_{\text{quality}}(\nu)$$

is not a simple product because F_{quality} is itself a declining function of ν. As ν increases, the shorter site lifetimes suppress consolidation, eroding the field before births can relay it. This creates a feedback loop: higher ν lowers F_{quality}, which weakens copy-forward, which further reduces F_{quality}.

3.3 Two Regimes of Structural Maintenance

Together, Experiments 1 and 2 resolve the copy-forward pathway's role:

- **Low ν ($\nu \lesssim 5 \times 10^{-4}$)**: Long site lifetimes allow consolidation to build high F_{quality}. Copy-forward is irrelevant because births are rare. ν^* and sg4 are nearly identical whether $\beta_s = 0$ or $\beta_s = 0.25$.
- **Moderate ν ($5 \times 10^{-4} < \nu \lesssim 0.005$)**: Sites die faster than the calm-streak gate allows consolidation. Copy-forward becomes the dominant maintenance pathway: each birth reinjects neighbor field structure. β_s now strongly controls both sg4 height and ν^*.
- **High ν ($\nu > 0.005$)**: Even copy-forward fails. Field quality $F_{\text{quality}} \to 0$ because consolidation cannot keep up at all. Structure collapses regardless of β_s.

This defines *three turnover regimes* rather than two (frozen/adaptive): consolidation-dominated, copy-forward-dominated, and collapsed. The adaptive window of prior papers corresponds to the union of the first two regimes.

4 Discussion

4.1 The Control Space is Four-Dimensional

Papers 8–12 have now identified four independent axes of the adaptive window:

1. **Turnover** ν: the external control parameter.
2. **Memory decay** (FIELD_DECAY $\to \nu_{\text{cryst}}$): sets lower boundary independently of all other parameters.
3. **Consolidation access** (SS, α_F, WR/WD $\to P_{\text{calm}}$, $\nu_{\text{max}}^{\text{consol}}$): sets upper consolidation boundary.
4. **Copy-forward** ($\beta_s \to \nu_{\text{max}}^{\text{cf}}$): extends the upper boundary into the moderate-turnover regime, independently of P_{calm}.

These four axes are mutually irreducible: no two-dimensional scalar (Ξ, Ψ, or product $\nu\beta_s$) collapses the data across all experiments. The adaptive regime is better described as a *viability volume* in this four-dimensional control space:

$$\mathcal{V} = \left\{ (\nu,\, \text{FD},\, P_{\text{calm}} \cdot \alpha_F,\, \beta_s) : \nu_{\text{cryst}} < \nu < \max(\nu_{\text{max}}^{\text{consol}},\, \nu_{\text{max}}^{\text{cf}}) \right\}$$

The prior series of partial scalars were projections of this volume onto two-dimensional slices: Ξ captured the consolidation face (Papers 8–9); Ψ captured the consolidation–decay trade-off (Paper 10); the substrate constants paper (Paper 11) located the volume boundaries; this paper (Paper 12) maps the copy-forward face.

4.2 Why the 2D Attempts Partly Worked

Each 2D projection succeeded within its constrained parameter range and failed at the boundaries:

- Ψ predicted the window for weak β_s (low copy-forward contribution) but failed at high WR and extreme SS (where copy-forward anomalies dominated).
- The SS sweep (Paper 11) predicted window collapse at SS= 40 but found persistent structure; copy-forward maintained sg4=45 even with $P_{\text{calm}} \approx 10^{-4}$.

These are not failures of the parameter sweep design; they are geometrically necessary consequences of projecting a 4D object onto a 2D face.

4.3 Implications for Paper 13

The appropriate goal for Paper 13 is not a single universal scalar but a characterization of the minimal state space sufficient to predict ν^* and sg4*. We propose to test whether a 4D inequality system:

$$\nu > \nu_{\text{cryst}}(\text{FD}), \qquad \nu < \nu_{\text{max}}^{\text{consol}}(P_{\text{calm}},\, \alpha_F) + \nu_{\text{max}}^{\text{cf}}(\beta_s,\, F_{\text{quality}}(\nu))$$

with $\nu^* \approx \sqrt{\nu_{\text{cryst}} \cdot \nu_{\text{max}}^{\text{total}}}$ as the geometric mean, suffices to predict ν^* across the full parameter range of Papers 8–12.

5 Conclusion

The copy-forward pathway ($\beta_s > 0$) is a fourth independent axis of the adaptive window. It is redundant with consolidation at low ν but becomes the primary structural maintenance mechanism at moderate turnover. The simple product $\nu\beta_s$ does not collapse the data because field quality F_{quality} degrades nonlinearly with ν, creating a feedback loop. The adaptive regime of VCML is governed by a four-dimensional viability volume, not a single scalar, and each paper in this series has mapped one face of that object. Paper 13 characterizes the complete volume.

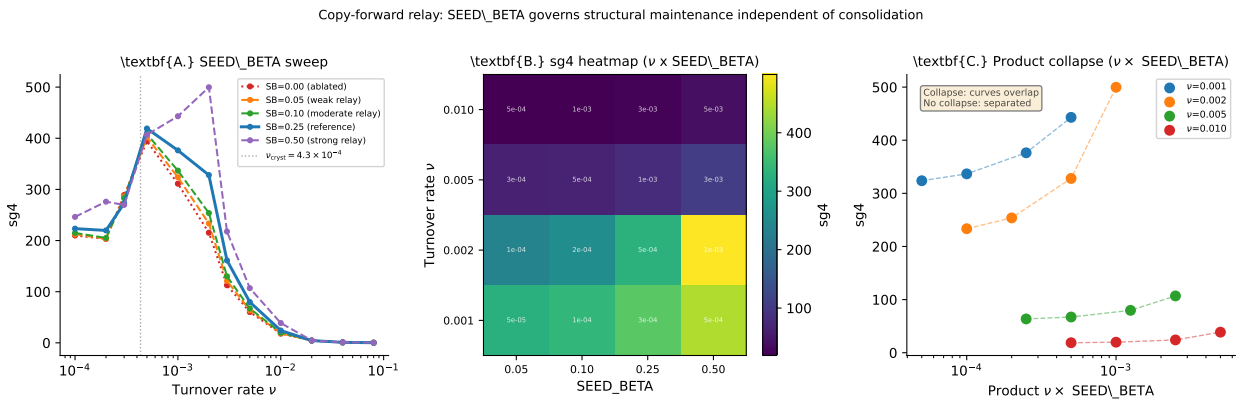

Figure 1: **A.** SEED_BETA sweep across the full ν range. At low ν the profiles are nearly identical (consolidation dominates); at $\nu = 0.002$, sg4 doubles from $\beta_s = 0$ to $\beta_s = 0.50$. Strong copy-forward ($\beta_s = 0.50$) shifts ν^* from 5×10^{-4} to 2×10^{-3}. **B.** sg4 heatmap over the $\nu \times \beta_s$ grid. Diagonals correspond to iso-product pairs; their large variation in sg4 falsifies the product hypothesis. Products (italicized values) span the same range but sg4 varies 24×. **C.** Scatter of sg4 vs. $\nu\beta_s$ colored by ν. Separated curves confirm that ν and β_s have independent effects beyond their product.